

Methodology/ delivery approach

Design Science Research (DSR) will be used as the primary methodology of this project while Agile Scrum will be adopted for delivering the approaches iteratively. DSR is appropriate because the main output of this project will be the design, build and testing of an AI-enabled artefact, the Generative AI Career Readiness Assistant for International Students in Australia. Besides other theory-based research, DSR uses a formalized process for developing new IT artefacts that address real world problems with both systematic design and evaluation (Hevner, 2007; Hevner and Chatterjee, 2010; Peffers et al., 2007).

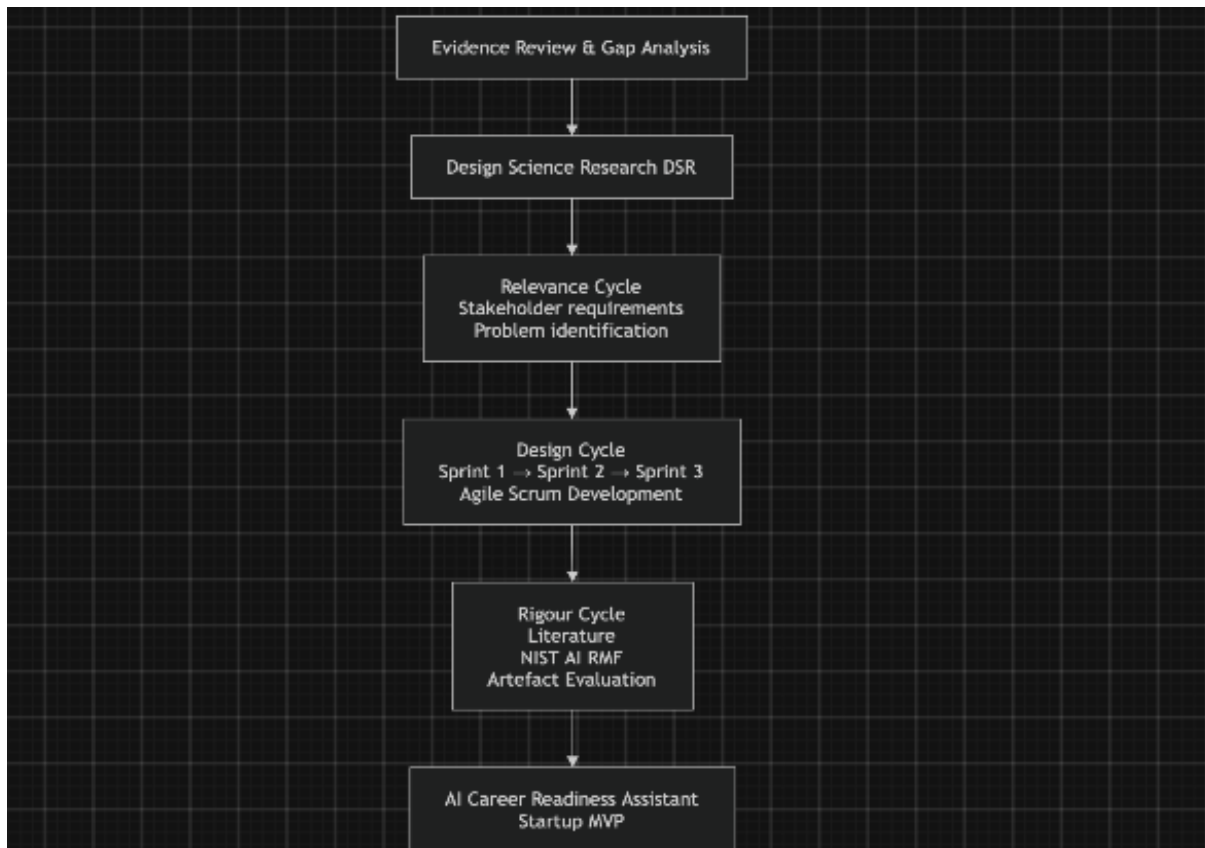


Figure 2: Proposed Methodology and Delivery Approach

Figure 2 illustrates the proposed methodology and delivery approach. It shows how Design Science Research (DSR) guides the overall project while Agile Scrum supports the overall design, development and testing of the proposed Startup MVP. As shown in Figure 2, the project uses Hevner's Three DSR Cycles. The Relevance Cycle is based on the evidence review, stakeholder's needs and gap analysis from assessments 1 and 2. The assessment found that there was a lack of effective integration among current AI Career support services. The Design Cycle provides guidance for the iterative development, testing and improvement of this Startup MVP through Agile Scrum sprints (Hevner et al., 2004). The Rigour Cycle will ensure that the artefact is developed with well-established research and industry frameworks such as Generative AI and NIST AI RMF to ensure transparency and fairness (Dwivedi et al., 2023; NIST, 2023).

The project has adopted an Agile Scrum methodology for the delivery of the project (Schwaber and Sutherland, 2020) to ensure successful implementation. The MVP will be built in three iterative sprints, where each sprint will provide functional enhancements to the product which will then be evaluated against the projects' objectives before proceeding to the next phase. This method of

iteration will allow for continued refining of the platform during development and to incorporate the input from both stakeholders and supervisors continuously throughout the development process (Schwaber and Sutherland, 2020).

The preference of DSR over other methodologies is based upon the fact that the main deliverable of this project is a working functional AI prototype and therefore not theory-based outcomes. The preference for DSR over utilizing Agile scrum exclusively is due to the fact that although the iterative software development process of Agile scrum is essential for the project's overall success as it is equally important that there is systematic evaluation of whether or not the artefacts produced addresses the identified employability gap. Therefore, Agile Scrum provides flexibility through its delivery framework to accommodate both incremental MVP building and continuing improvement throughout all phases of the project life cycle.